

Math 242, S19
Quiz 3

Name: Answer

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Find the particular solution of the initial value problem:

$$x \frac{dy}{dx} = 2y + x^3 \sin x \cos x, \quad y(\pi) = 2.$$

(First transform the differential equation into the standard form of linear first-order differential equation if it is not.)

Transform to its standard form:

$$\frac{dy}{dx} - \frac{2}{x}y = x^2 \sin x \cos x$$

$$P(x) = -\frac{2}{x}, \quad Q(x) = x^2 \sin x \cos x$$

Thus the integrating factor

$$S(x) = e^{\int P(x) dx} = e^{\int -\frac{2}{x} dx} = e^{-2 \ln|x|} = \frac{1}{x^2}$$

By multiplying $S(x)$ to both sides of the equation,
we get

$$\frac{d}{dx} \left[\left(\frac{1}{x^2} \right) y \right] = (x^2 \sin x \cos x) \left(\frac{1}{x^2} \right)$$

$$\frac{y}{x^2} = \int \sin x \cos x dx$$

$$\frac{y}{x^2} = \frac{1}{2} \sin^2 x + C$$

$$\Rightarrow y = x^2 \left(\frac{1}{2} \sin^2 x + C \right)$$

$$\begin{aligned} y(\pi) &= 2 \\ \Rightarrow \pi^2 \left(\frac{1}{2} \sin^2 \pi + C \right) &= 2 \\ \Rightarrow C &= \frac{2}{\pi^2} \end{aligned}$$

Thus we have

$$\underline{y = x^2 \left(\frac{1}{2} \sin^2 x + \frac{2}{\pi^2} \right)}$$